Mock Final Exam - Version V5

Total Points: 100

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Multiple Choice (12 questions, 3 points each = 36 points)

Select the best answer for each question.

1. (3 points) The primary advantage of using binary I/O over text I/O is often:

A. Human readability.

B. Easier debugging.

C. Platform independence for character encoding.

D. Efficiency in storage space and processing speed for non-string data.

2. (3 points) Which class allows you to read and write objects directly to/from a stream, preserving their state?

A. DataInputStream / DataOutputStream

B. FileInputStream / FileOutputStream

C. ObjectInputStream / ObjectOutputStream

D. RandomAccessFile

3. (3 points) What is the purpose of the getFilePointer() method in RandomAccessFile?

A. To set the position where the next read/write will occur.

B. To get the total length of the file in bytes.

C. To get the current position (offset in bytes from the start) of the file pointer.

D. To check if the file pointer is at the end of the file.

4. (3 points) Writing fixed-length records to a RandomAccessFile is useful because it:

A. Saves disk space compared to variable-length records.

B. Allows easy calculation of the starting position of any record using record_index * record_size.

C. Is the only way to store String data in a RandomAccessFile.

D. Automatically handles EOFException.

5. (3 points) Which of these problems is LEAST likely to be solved elegantly using recursion?

A. Traversing a file system directory structure.

B. Calculating the Nth Fibonacci number.

C. Implementing merge sort.

D. Reading records sequentially from a binary file using a simple loop.

6. (3 points) Consider the method mystery(n) defined in Mock Exam V2, Question 14. What does this method effectively calculate for a non-negative integer n?

A. The sum of all integers from 1 to n.

B. The sum of only the even integers from 1 to n.

C. The factorial of n.

D. The nth Fibonacci number.

7. (3 points) If you implement a recursive binary search on a sorted array, the recursive step typically involves:

A. Comparing the key with the first element and recursing on the rest.

B. Comparing the key with the middle element and recursing on either the left or right half.

C. Swapping elements to bring the key closer to the beginning.

D. Iterating through the array elements recursively.

8. (3 points) In JavaFX, what is the relationship between a Stage and a Scene?

A. A Stage contains one or more Scenes.

B. A Scene contains one or more Stages.

C. A Stage is the root node within a Scene.

D. A Stage displays a single Scene at a time.

9. (3 points) Which JavaFX event handling technique is generally preferred for simple, single-use handlers due to its conciseness?

A. Separate top-level handler classes.

B. Using the controller class as the handler.

C. Named inner classes.

D. Lambda expressions.

10. (3 points) If class B extends class A, and both have a method m(), how does Java determine which version of m() to run when called on an object of type B referred to by a variable of type A?

A. It always runs the version in A (static binding).

B. It runs the version in B (dynamic binding/polymorphism).

C. It depends on the access modifier of the methods.

D. It results in a compile-time error.

11. (3 points) What happens if a class implements an interface but fails to provide an implementation for one of the interface's abstract methods?

A. The program runs, but calls to that method will fail.

B. The interface provides a default implementation automatically.

C. The class must be declared abstract itself.

D. A runtime exception MissingMethodException occurs.

12. (3 points) Text file I/O (using FileReader/PrintWriter) is generally better suited than binary I/O for:

A. Storing large arrays of double values efficiently.

B. Saving the state of complex Java objects.

C. Creating configuration files or logs meant to be easily read/edited by humans.

D. High-performance data transfer between applications.

Short Answer / Code Reading

1. (4 questions, variable points = 29 points)

2. (6 points) When writing fixed-length strings to a RandomAccessFile, why is padding often necessary? How might you implement reading and writing such a fixed-length string (e.g., always 20 characters)?

3. (7 points) Look at the countDown method from Mock Exam V2, Question 14. Is this method tail-recursive? Explain why or why not.

4. (8 points) You have an ArrayList<Shape> where Shape is an abstract class with subclasses Circle and Rectangle. Explain how polymorphism allows you to iterate through this list and call an abstract getArea() method on each element, automatically executing the correct area calculation for either the Circle or Rectangle object stored there.

5. (8 points) Describe the purpose of the try-with-resources statement in Java I/O. What common task does it automate, and what kind of resource classes can be used with it?

Coding / Program Design

1. (4 questions, variable points = 35 points)

2. (10 points) Write a recursive Java method binarySearchRec(int[] sortedArr, int key, int low, int high) that performs a binary search for key within the sorted integer array sortedArr between indices low and high (inclusive). It should return the index of key if found, or -1 if not found. You must use recursion. Define the base case(s) and recursive step clearly within your code using comments.

```
public static int binarySearchRec(int[] sortedArr, int key, int low, int high) {
// Base Case(s): ?? (found or not found)
// Recursive Step: ?? (check middle, recurse left or right)
}
```

3. (12 points) Assume you have the InventoryItem class from Mock Exam V2, Question 18 (which is Serializable). Write a Java code snippet that:

- Reads the ArrayList of InventoryItem objects from the binary file "inventory.dat" using ObjectInputStream (wrapped appropriately).
- Iterates through the loaded ArrayList.
- For each InventoryItem whose itemCount is less than 10, print its name and count to the console.
- Include exception handling for IOException, ClassNotFoundException, and EOFException.

4. (13 points) Create a JavaFX application snippet within the start(Stage primaryStage) method that:

- Creates a VBox layout pane.
- Creates a Label with the text "Enter Name:".
- Creates a TextField.
- Creates a Button with the text "Submit".

- Adds the Label, TextField, and Button to the VBox.

- Sets up an ActionEvent handler (using a lambda expression) for the Button. When clicked, the handler should retrieve the text from the TextField and print "Hello, [entered text]!" to the console.

- Creates a Scene with the VBox as its root.

- Sets the scene onto the primaryStage and shows the stage.